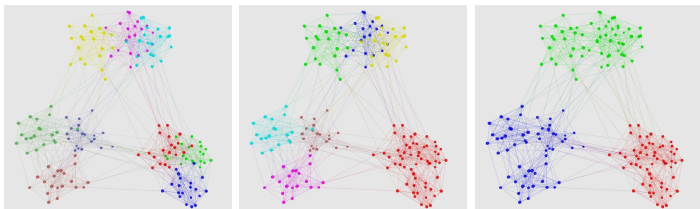


User Guide

Navigate through a graph clustered
with several scales of description

May, 06, 2026



Presentation

Networks are one of the main conceptual structures for modeling complex systems, where nodes represent the basic lowest-level entities of the system and edges represent their interactions. Graph clustering algorithms are then used to identify modules : over-densely connected sets of nodes of these networks as representations of entities at higher scales of description in these systems.

bec is a C++ graph clustering program that finds the entities of a network with a hand on the **description scales** within the **nPnB framework**.

■ **nPnB** : B. GAUME, I. ACHITOUV, AND D. CHAVALARIAS : *Two antagonistic objectives for one multi-scale graph clustering framework* Nature Scientific Reports, 15 (2025), p. 13368. <https://hal.science/hal-05046050v1>

■ **bec** : B. GAUME : *Find Optimal clusterings in nPnB framework is hard, BEC.2 : an algorithm to find high-performing ones* Forthcoming (2026)

■ **description scales** : B. GAUME : *Defining the appropriate scales for describing a complex system* Forthcoming (2026)

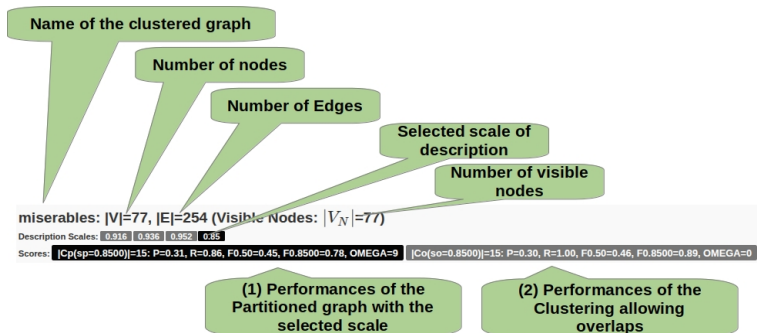
Presentation

In the following, we describe how to navigate through graphs clustered with several scales of description.

The screenshot shows a complex graph interface with several callouts explaining its features:

- Information on the graph and its clusterings**: A green callout pointing to the top status bar which displays graph statistics: `miserables: |V|=77, |E|=254 (Visible Nodes: |Vv|=77)` and various parameters like `Scale: [Cp=0.8000]-15, P=0.31, R=0.85, FS=50-0.45, FS=800-0.75, OMEGA=0`.
- Button Focus**: A blue callout pointing to the 'Focus' button in the top left.
- Button Rotation**: A blue callout pointing to the 'Rotation' button in the top left.
- Selects a node**: A blue callout pointing to the 'Labels' input field in the top left.
- Selects Nodes-Visibility From one node**: A blue callout pointing to the 'UPDATE NODES-VISIBILITY FROM: 01: Garroche' menu.
- Information on the node**: A green callout pointing to the detailed node information panel for '01: Garroche (in module 6)', showing its neighbors and visible neighbors.
- Button Export**: A blue callout pointing to the 'EXPORT' button in the top right.
- Selects Nodes-Visibility**: A blue callout pointing to the 'ON DEGREES' and 'ON LABELS' options in the bottom right panel.

Informations on the graph and its clusterings



- ▶ (1) selected $\Rightarrow$ nodes are colored with C_p
- ▶ (2) selected $\Rightarrow$ nodes are colored with C_o

Let m a set of nodes, then $\omega_G(m) = \left| \{i \in \overline{m} \text{ such } \forall j \in m, \{i, j\} \in E\} \right|$
For a clustering C on a graph G , $OMEGA(C) = \sum_{m \in C} \omega_G(m)$
 C is an intrinsic Clustering $\Leftrightarrow C$ is a node partition such $OMEGA(C) = 0$

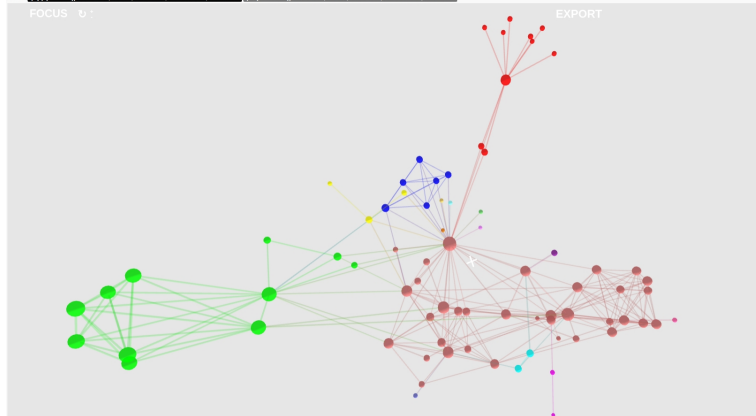
Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.85, selected clustering : C_p

miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.852 | 0.936 | 0.952 | 0.85

Scores: $[C_p]_{\alpha=0.8500}[-10: P=0.91, R=0.86, F0.50=0.45, F0.8500=0.76, \text{OMEGA}=9]$ $[C_0]_{\alpha=0.8500}[-10: P=0.30, R=1.00, F0.50=0.46, F0.8500=0.89, \text{OMEGA}=0]$



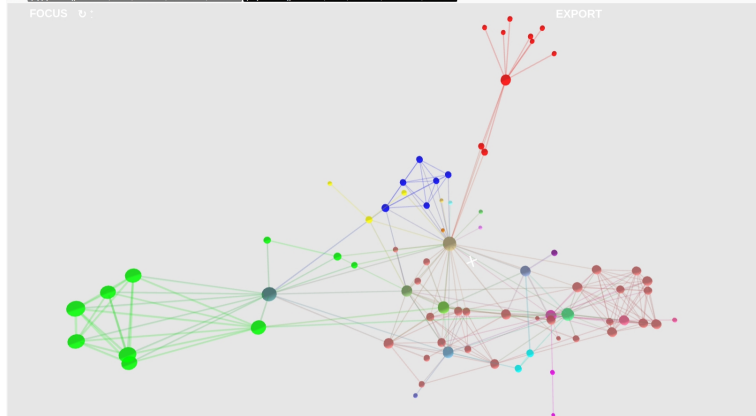
Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.85, selected clustering : Co

miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.858 | 0.936 | 0.952 | 0.85

Scores: $[Co(su)=0.8500][c=10: P=0.91, R=0.86, F0.50=0.45, F0.8500=0.76, OMEGA=9]$ $[Co(sa)=0.8500][c=10: P=0.30, R=1.00, F0.50=0.46, F0.8500=0.89, OMEGA=0]$



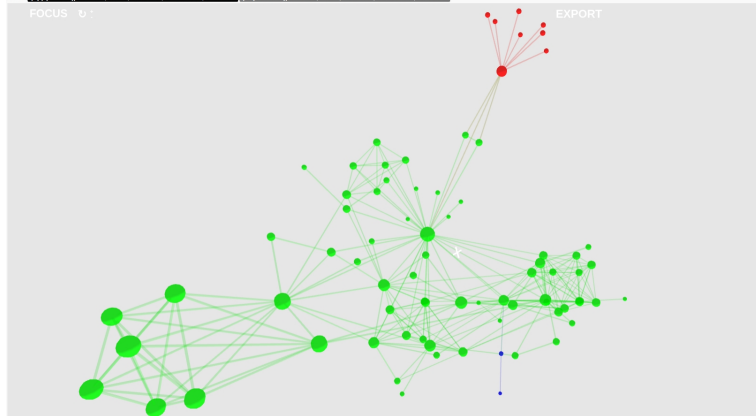
Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.916, selected clustering : C_p

miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.916 | 0.936 | 0.952 | 0.967

Scores: $[C_p(\alpha)=0.9160][\alpha=0.916, R=0.98, F0.50=0.29, F0.9166=0.87, \text{OMEGA}=0]$ $[C_0(\alpha)=0.9160][\alpha=0.916, R=0.98, F0.50=0.29, F0.9166=0.88, \text{OMEGA}=0]$



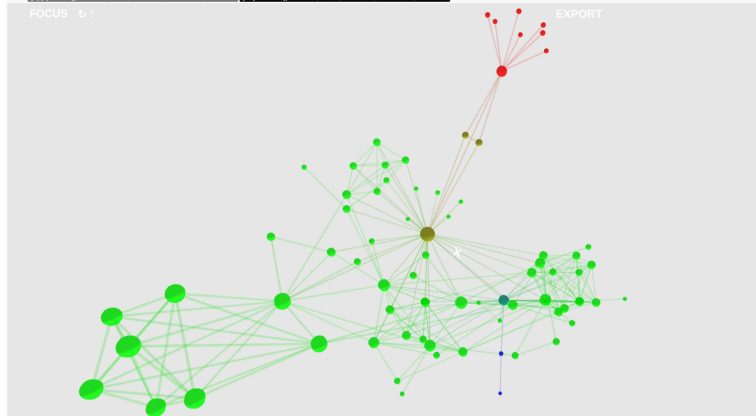
Informations on the graph and its clusterings

EXAMPLE : selected scale : 0.916, selected clustering : Co

miserables: $|V|=77$, $|E|=254$ (Visible Nodes: $|V_N|=77$)

Description Scales: 0.916 | 0.936 | 0.952 | 0.960

Scores: $[Co(su=0.9160)] = 3$; $P=0.11$, $R=1.00$, $F0.50=0.20$, $F0.9160=0.88$, $OMEGA=0$



Informations on a node

Hovering over a node : displays its modules and neighbors

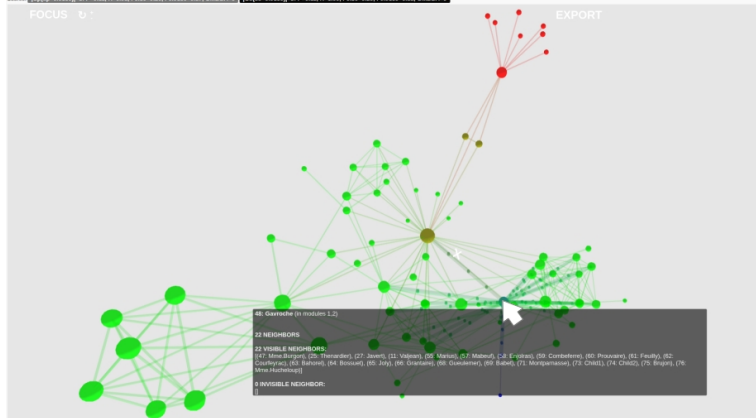
miserables: |V|=77, |E|=254 (Visible Nodes: $|V_N|=77$)

Description Scores: 0.512 0.336 0.02 0.00

Scores: |Cajon|=0.9160|c|=3, P=0.11, R=0.46, F0.50=0.26, F0.9160=0.87, OMEGA=0 |Cajon|=0.9160|c|=3, P=0.11, R=1.00, F0.50=0.20, F0.9160=0.86, OMEGA=0

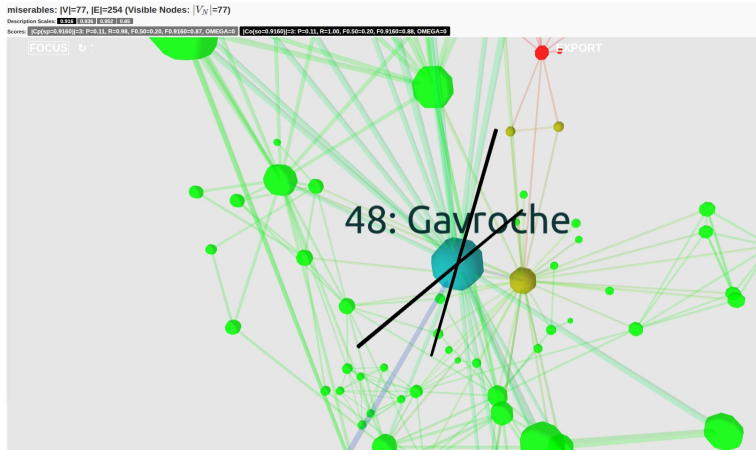
FOCUS u :

EXPORT



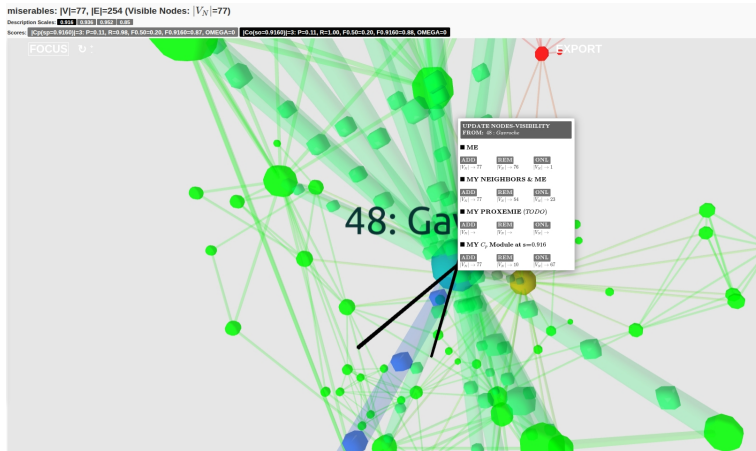
Informations on a node

Left-click on the node : zoom on it



Selects Nodes-Visibility From one node

Rigth-click on the node : Update nodes-visibility from this node



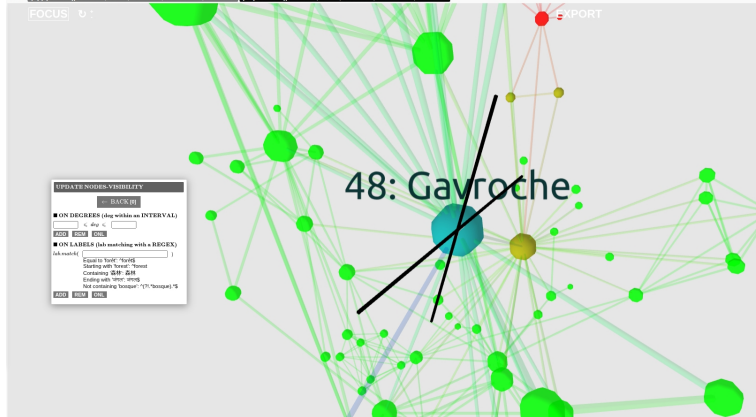
Update nodes-visibility From on their properties

Right-click on empty space : Update nodes-visibility from their properties

miserables: |V|=77, |E|=254 (Visible Nodes: |V_N|=77)

Description Scores: 0.814 | 0.836 | 0.852 | 0.865

Scores: |Cq(su)=0.9160|=-3; P=0.11, R=0.48, F0.50=0.29, F0.9160=0.87, OMEGA=0 |Cq(su)=0.9160|=-3; P=0.11, R=1.00, F0.50=0.20, F0.9160=0.88, OMEGA=0



FOCUS Button

Right-click on FOCUS button : Search for a node by its label or index

